

## SUMMARY TABLE: PHYSICS GRADE 10

### Sub-Strand 1.2: Mechanical Properties of Materials — Teacher Reference (Pre-filled)

| SUMMARY TABLE: PHYSICS GRADE 10 |                                                                                                                                                                                                        |
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| Sub-Strand                      | Sub-Strand 1.2: Mechanical Properties of Materials                                                                                                                                                     |
| Driving Question                | DRIVING QUESTION: Why do some materials stretch and snap back while others permanently deform or break — and how can we use this knowledge to choose the right material for a safe structure in Kenya? |

### INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

| Lesson # and Title                                                                                                                                    | What did I observe?                                                                                                                                                                                                                                                                                                                                                                                                                              | What did I learn?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | How does this explain the phenomenon?                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <b>Lesson 1</b><br><b>Predict and Observe:</b><br><b>Elastic vs Plastic</b><br><b>Behaviour — The</b><br><b>Kericho Bridge Rope</b><br><b>Mystery</b> | One sisal rope stretched and returned to its original shape when the load was removed, while a second sisal rope stretched permanently and then snapped under the same 80 kg load. In the hands-on exploration, rubber bands returned to their original length (elastic behaviour), sisal twine showed slight permanent deformation (borderline plastic), and copper wire remained bent after the load was removed (plastic behaviour).          | Elastic behaviour describes a material that returns to its original shape and size after the applied force is removed. Plastic deformation describes a material that remains permanently changed in shape after the force is removed. These behaviours can be explained at the particle level: in elastic materials, particle bonds stretch and return; in plastic materials, particles shift into new permanent positions. Both the Kericho ropes appeared similar on the outside, but their internal structure and condition determined their behaviour under load. | The new sisal rope (rope A) behaved elastically because its particle bonds could stretch and recover under the 80 kg load. The old sisal rope (rope B) underwent plastic deformation and eventually fractured because its internal structure had degraded — reducing its ability to recover from stretching. This introduces the key question for the unit: what specific measurable properties determine whether a material will behave elastically or plastically under a given force? |
| <b>Lesson 2</b><br><b>Stretching the Truth:</b><br><b>Elastic vs. Plastic</b><br><b>Behaviour in Real</b><br><b>Materials</b>                         | Learners stretched rubber bands, sisal twine, and copper wire under increasing loads (100 g to 500 g), recording extension and whether each material returned to its original length after the load was removed. They observed that the rubber band consistently returned to its original length across all loads tested, the sisal twine returned at low loads but showed permanent lengthening beyond approximately 300 g, and the copper wire | Learners established definitions for elasticity (a material's ability to return to its original shape after a deforming force is removed) and plasticity (permanent deformation after a deforming force is removed). They identified the elastic limit as the critical boundary between these two behaviours, and connected these concepts to the Kericho bridge phenomenon by classifying the new rope as elastic and the old, fatigued rope as having exceeded its                                                                                                  | Using a particle-level model, learners began to explain elastic and plastic behaviour in terms of inter-particle bonds: in elastic materials, applied force stretches inter-particle bonds but does not break them, so particles return to their original positions when the force is removed; in plastic materials, the applied force displaces particles beyond the range of their restoring bonds, leaving them in new permanent positions. This model is initial and                 |

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|                                                                                                                          | permanently bent after even a small load of 100–200 g.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | elastic limit and behaving plastically before snapping.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | qualitative — it will be refined and made quantitative in Lessons 3 and 4 when Hooke's Law and stress-strain graphs are introduced.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Lesson 3</b><br><b>Elastic Potential Energy and the Elastic Limit: How Much Can a Material Store Before It Fails?</b> | Force-extension graphs from Lesson 2 showing the linear (Hooke's Law) region and the point of elastic limit for the steel spring; the triangular area under the force-extension curve shaded by the teacher; the two overlapping force-extension curves representing the new and older Kericho sisal ropes, with different gradients and different elastic limits marked.                                                                                                                                               | Elastic potential energy is stored in a stretched material and equals $\frac{1}{2}kx^2$ , which is equivalent to the area of the triangle under the force-extension graph in the linear region; the elastic limit is the boundary beyond which a material cannot return to its original shape and may fail permanently; the safe working load is the maximum load a material can bear while remaining within its elastic limit; the older Kericho rope failed because the load exceeded its safe working load, pushing it past its elastic limit and causing permanent deformation and snap.                                                                 | The older Kericho sisal rope snapped because the weight of the tea sack (approximately 800 N) exceeded the rope's safe working load — that is, the force corresponding to the rope's elastic limit. Beyond that limit, the material could no longer store and return energy elastically; instead it deformed plastically and ultimately fractured. The new rope, with a higher elastic limit and greater spring constant, remained within its safe working load under the same force and so recovered elastically when the load was removed. This explanation is now supported by quantitative calculation of elastic potential energy and direct reading of the elastic limit from the force-extension graph.                                                                                                                              |
| <b>Lesson 4</b><br><b>Hooke's Law — Measuring Spring Constant and Finding the Elastic Limit</b>                          | Learners added slotted masses (0–5 N) to a hanging steel spring and measured the extension at each load increment using a metre rule clamped beside the retort stand. Half the class used a stiff spring; half used a soft spring. Learners recorded data in a force-extension table, plotted their results on graph paper, drew a line of best fit through the linear region, and marked the point where the graph began to curve (the elastic limit). They compared $k$ values across groups using a class data wall. | Learners learned that: (1) For a spring loaded within its elastic limit, extension is directly proportional to the applied force — this is Hooke's Law ( $F = kx$ ). (2) The spring constant $k$ (N/m) is the gradient of the linear portion of the force-extension graph; a larger $k$ means a stiffer spring requiring more force per unit extension. (3) Beyond the elastic limit, the spring no longer returns to its original length when the load is removed — permanent deformation begins. (4) Different springs have different $k$ values and different elastic limits, meaning their behaviour under the same force is quantitatively predictable. | Learners explained that the Kericho bridge rope that snapped had a lower spring constant and a lower elastic limit than the rope that survived. When the 80 kg tea-leaf sack ( $\approx 800$ N) was placed on the bridge, the force applied to the older rope exceeded its elastic limit, causing permanent deformation in the rope fibres. Because the force continued beyond the elastic limit toward the breaking point, the rope snapped. The rope that survived had a sufficiently high elastic limit that 800 N fell within its proportional region, allowing it to return to its original length when the load was removed. Learners revised their running rope models to show two force-extension graphs side by side, with a vertical line at 800 N demonstrating that Rope B's elastic limit was exceeded while Rope A's was not. |
| <b>Lesson 5</b><br><b>Stress, Strain, and the Young's Modulus:</b>                                                       | Learners loaded two rubber strips (representing new and old sisal rope) with increasing masses and measured their                                                                                                                                                                                                                                                                                                                                                                                                       | Stress ( $\sigma = F/A$ , measured in Pascals) describes how much force acts per unit area inside a material. Strain ( $\epsilon = \Delta L/L_0$ ,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | The farmer's old sisal rope failed because years of use, sun exposure, and repeated loading had degraded the rope's internal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

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| <b>Predicting Which Rope Will Snap</b>                                                                                                      | <p>extension at each step, recording force, cross-sectional area, original length, and extension in a structured data table. They observed that Strip B (fatigued/old rope) reached permanent deformation and fracture at a lower load than Strip A, and that the two strips produced visibly different stress-strain curves when plotted on the same axes.</p>                                                                                                                              | <p>dimensionless) describes how much a material deforms relative to its original length. Young's Modulus (<math>E = \sigma/\epsilon</math>) is a material constant that quantifies stiffness — a higher <math>E</math> means a stiffer material that requires more stress to produce the same strain. The stress-strain graph has five key regions: linear (Hookean) region, elastic limit, yield point, Ultimate Tensile Strength (UTS), and fracture point. The old Kericho rope snapped because its fracture stress was lower than the stress applied by the 80 kg tea-leaf sack — a fact that can be determined before loading by measuring <math>E</math> and comparing it to the applied stress.</p>                                                                    | <p>fibre structure, reducing its Young's Modulus and — critically — lowering its fracture stress below the stress imposed by the 80 kg load. The new rope survived because its higher Young's Modulus and higher fracture stress meant it could bear the applied stress while remaining in the elastic (recoverable) region. By measuring stress, strain, and Young's Modulus before selecting a rope, a farmer or engineer can predict — with numbers, not guesswork — which material is safe for a given load. This is the same principle used by Kenyan civil engineers when selecting cables, tendons, and support ropes for real bridges and structures.</p> |
| <b>Lesson 6</b><br><b>Final Explanation: Elastic Potential Energy, Elastic Limits, and Choosing the Right Material for a Safe Structure</b> | <p>Learners observed quantitative data showing that the old Kericho rope (<math>k = 210 \text{ N/m}</math>) was extended to 3.73 m under an 80 kg load — nearly double its elastic limit of 2.1 m — while the new rope (<math>k = 850 \text{ N/m}</math>) extended to only 0.92 m, well within its elastic limit. Learners observed parabolic <math>E_p</math> vs. extension graphs from their own spring data, and compared energy stored in two model ropes at the same applied force.</p> | <p>Learners learned that elastic potential energy stored in a stretched material is calculated using <math>E_p = \frac{1}{2}kx^2</math>; that the elastic limit is the maximum extension beyond which a material permanently deforms and cannot return to its original shape; that a higher spring constant means less extension under the same force and less energy stored; that the older Kericho rope failed because the applied load drove its extension far beyond its elastic limit, storing more energy than its molecular bonds could sustain; and that Hooke's Law and <math>E_p = \frac{1}{2}kx^2</math> can be applied quantitatively to real Kenyan engineering contexts such as matatu suspension springs and bungee cords to determine safe working loads.</p> | <p>Learners explained — using force-extension graphs, <math>E_p = \frac{1}{2}kx^2</math> calculations, and particle-level diagrams — why the older Kericho rope snapped while the newer rope survived the same load: the older rope had a much lower spring constant due to degraded fibres, causing it to extend far beyond its elastic limit, permanently disrupting molecular bonds and ultimately causing catastrophic failure. Learners applied this explanation to recommend a safe working load for the bridge and justify the selection of a higher-<math>k</math> replacement rope, directly answering the unit's driving question.</p>                  |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

**Production & Layout:** Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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